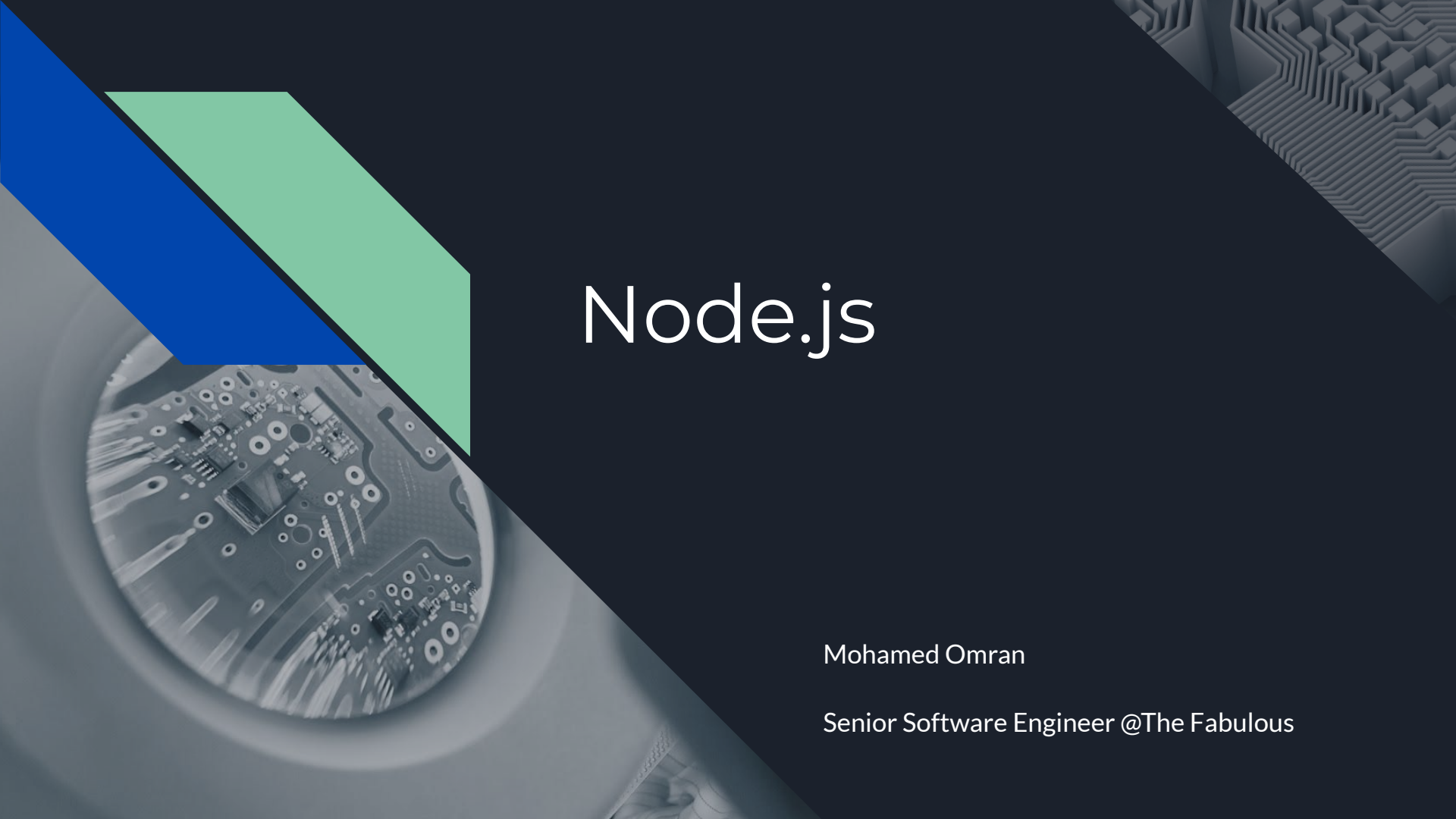




Node.js

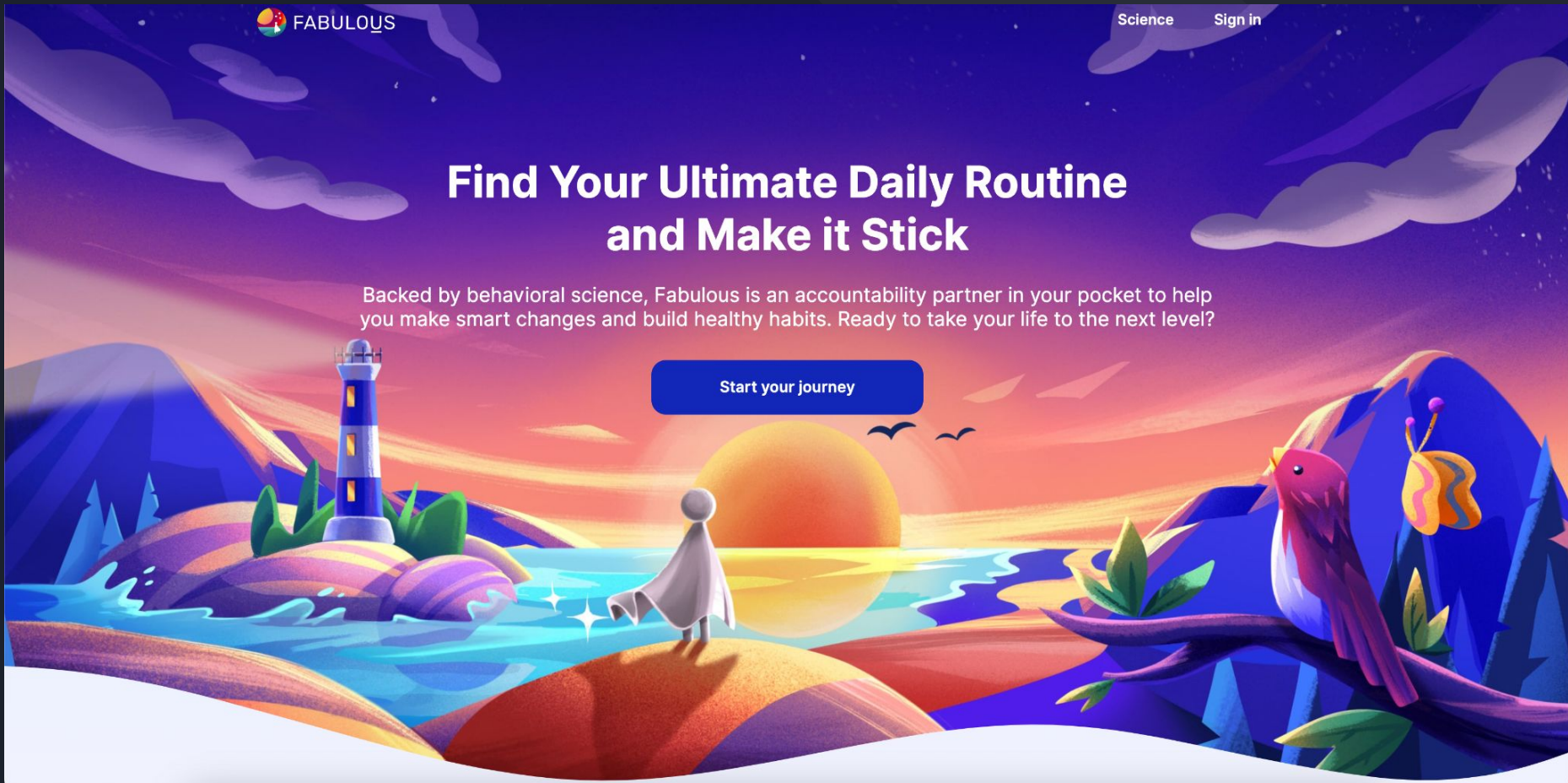
Mohamed Omran

Senior Software Engineer @The Fabulous

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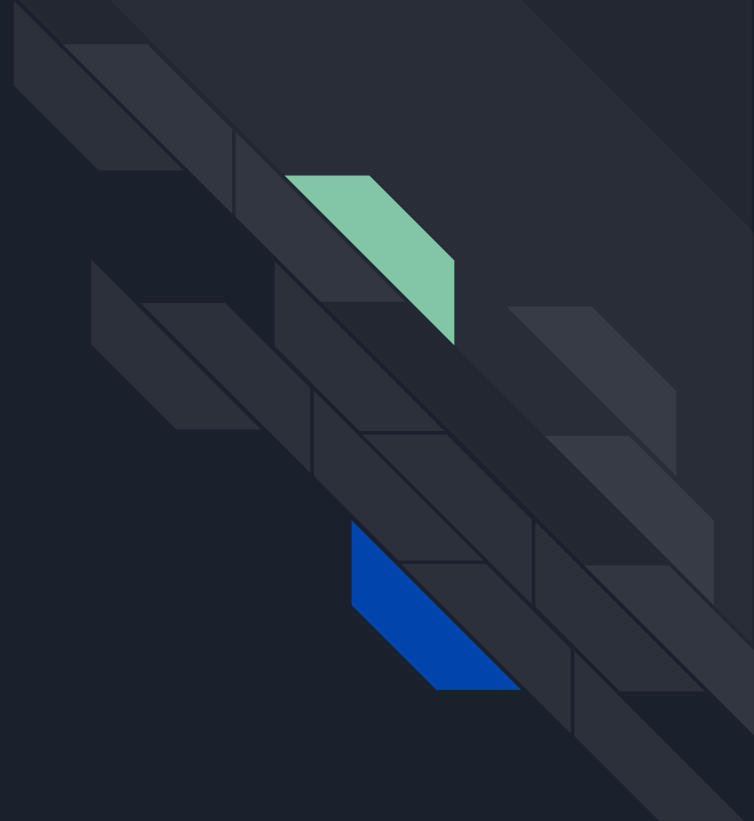
Start your journey





Questions

- Wait for the end of the section
- Try to keep your question in context
- Listen to others' questions
- Ask repeatedly until you understand



Content

Day 1

Web 101

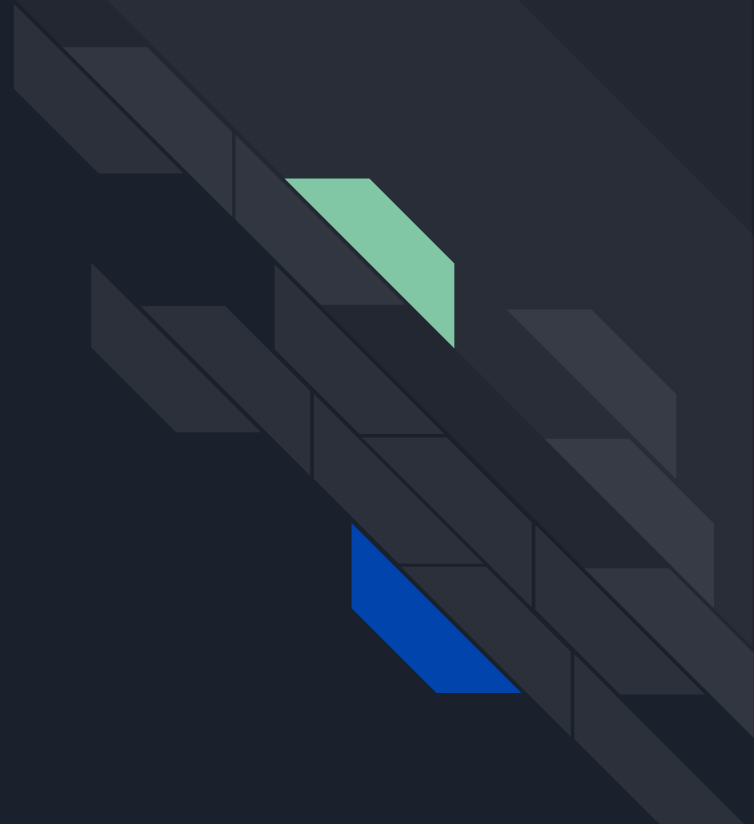
What is Node.js

Node.js vs Browser

ENV variable, global, REPL, CLI Args

Modules and require

Core Modules

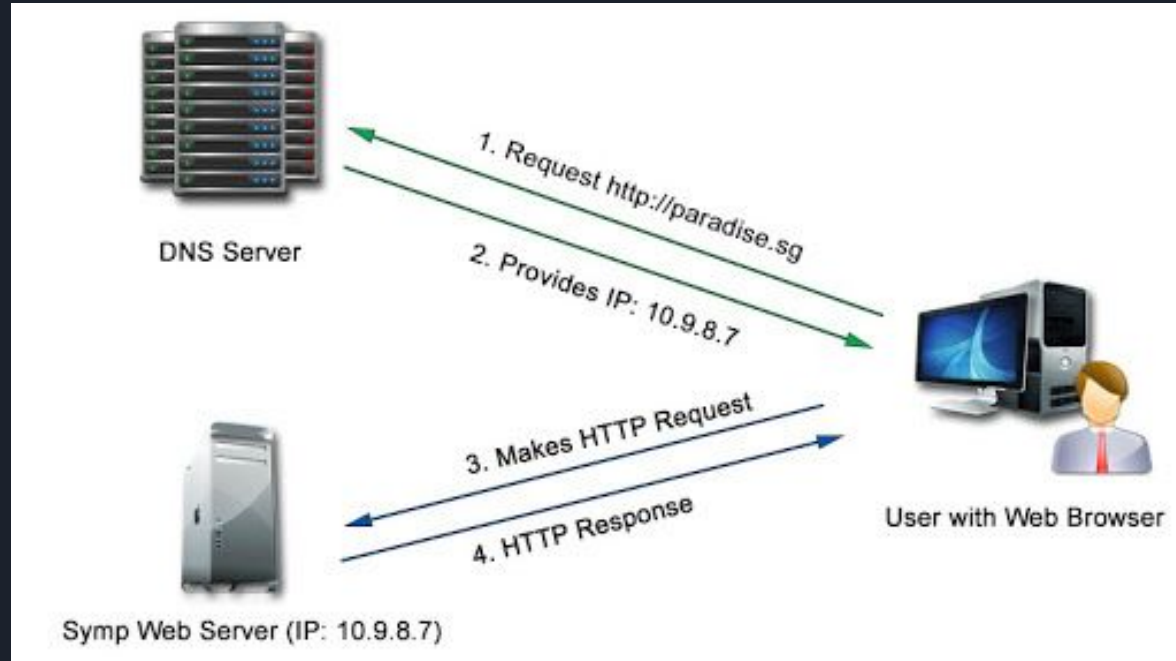




Web 101

- Web Server
- Protocols
- HTTP
 - Request
 - Response

Web 101



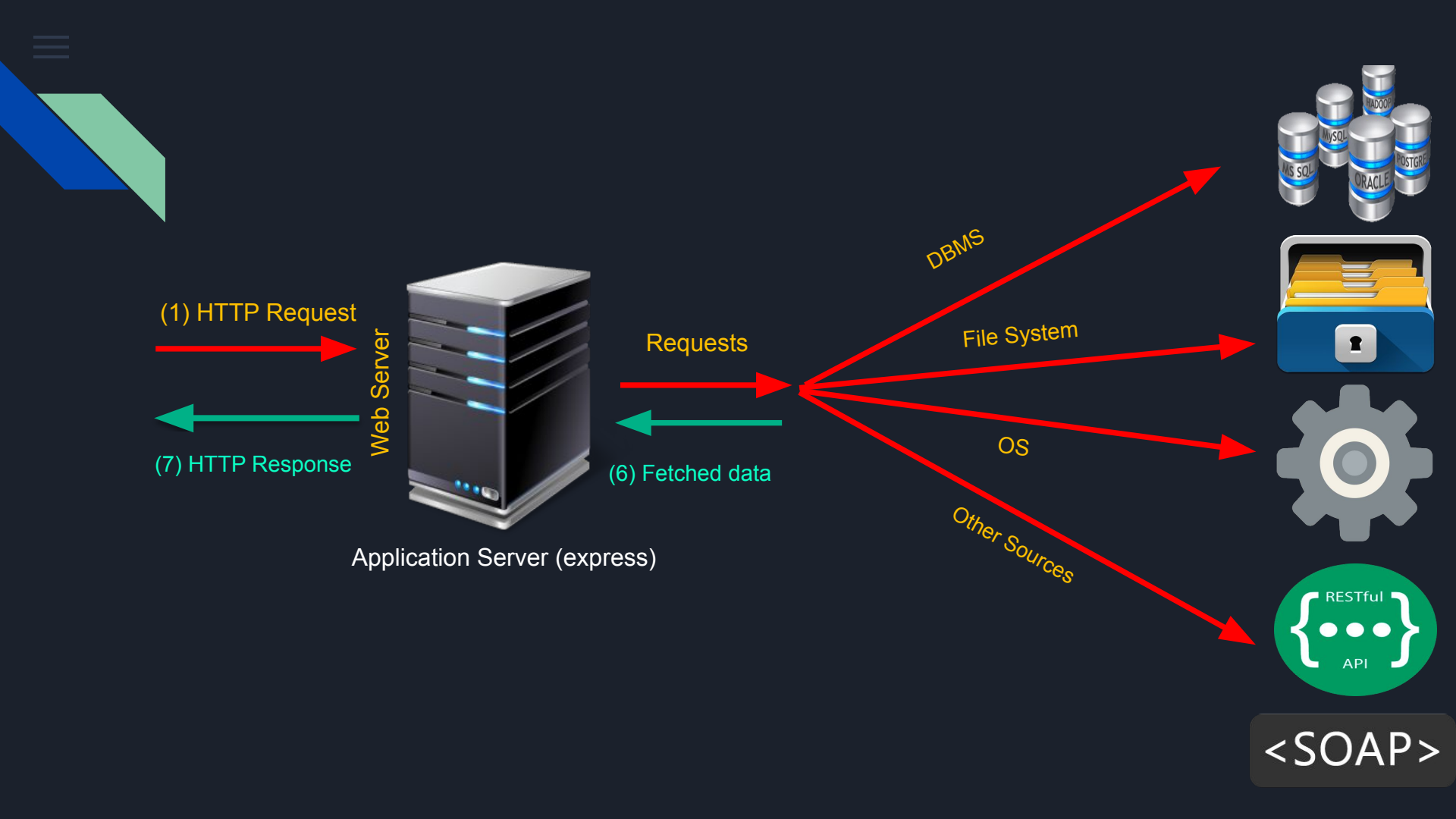
Web 101

HTTP URL Anatomy

1 2 3 4 5 6 7 8
<https://www.example.com:3000/path/resource?id=123#section-id>

Key

- 1 Scheme - defines how the resource will be obtained.
- 2 Subdomain - www is most common but not required.
- 3 Domain - unique value within its top-level domain.
- 4 Top-level Domain - hundreds of options now exist.
- 5 Port - if omitted HTTP will connect on port 80, HTTPS on 443.
- 6 Path - specify and perhaps find requested resource.
- 7 Query String - data passed to server-side software, if present.
- 8 Fragment Identifier - a specific place within an HTML document.





What's Node.js?



Node.js is an open-source, cross-platform, back-end, JavaScript runtime environment that executes JavaScript code outside a web browser.

Open source

Why Open Source Software (OSS)

Cost Reduction

Quality Improvement

Quick Time to Market

Full Ownership and control

Drive innovation with rapid pace

No vendor lock-in, great flexibility

Broad perspective (more eyes on the code)

Integration and Customization- Easy to modify and enhance

Collaboration approach gives better solutions- Community support



Requirements

- Deal with filesystem.
- Deal with network.
- Deal with DB.
- Accept requests and send responses.
- Way to deal with time consuming blocking tasks.
- Organize codebase.



Building a runtime

Application code (JavaScript)

Node.js bindings

C/C++ addons

V8

libuv

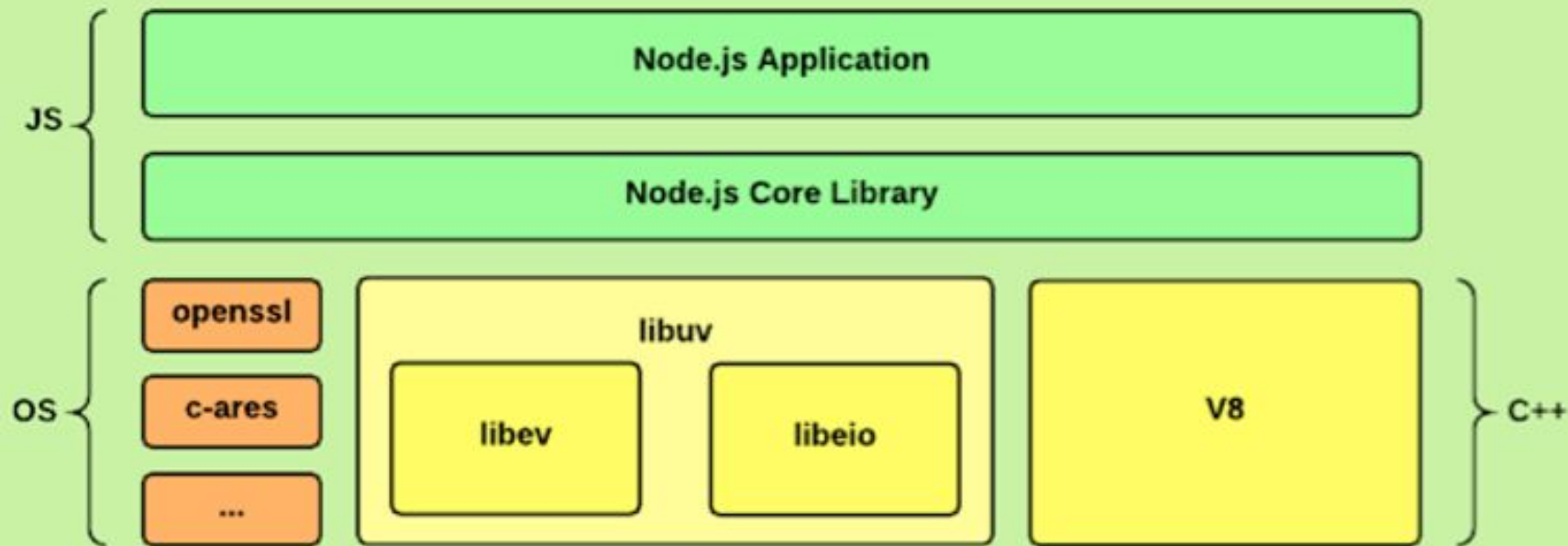
c-ares

http-
parser

OpenSSL

zlib

Building a runtime



Why using Node.js

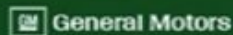
Node.js Advantages



NODE IS DEPLOYED BY BIG BRANDS

Big brands are using Node to power their business

Manufacturing



SIEMENS

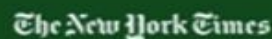
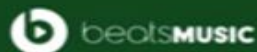
Financial



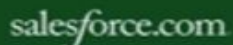
eCommerce



Media



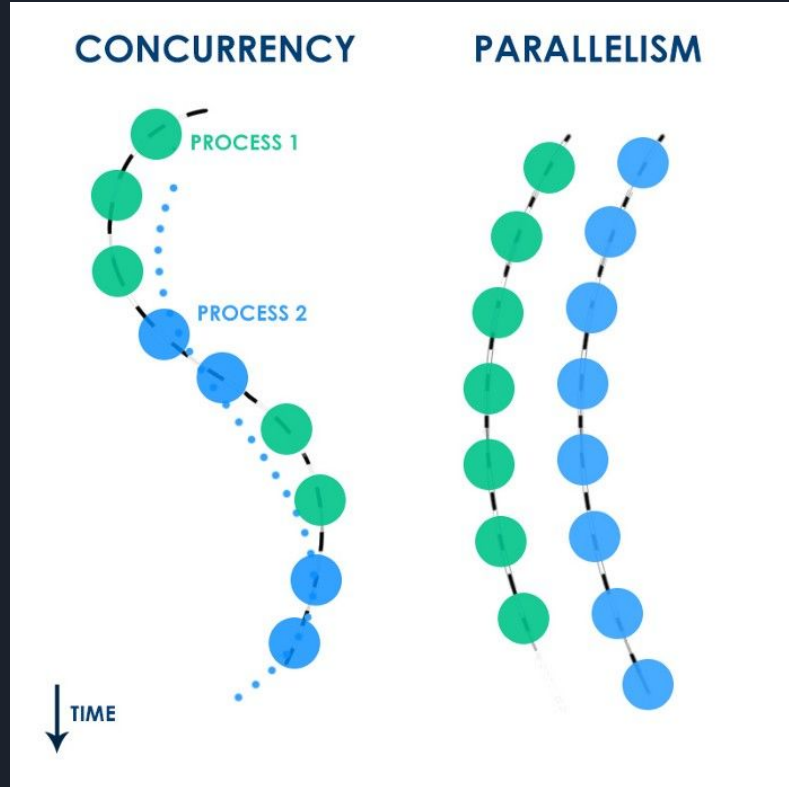
Technology



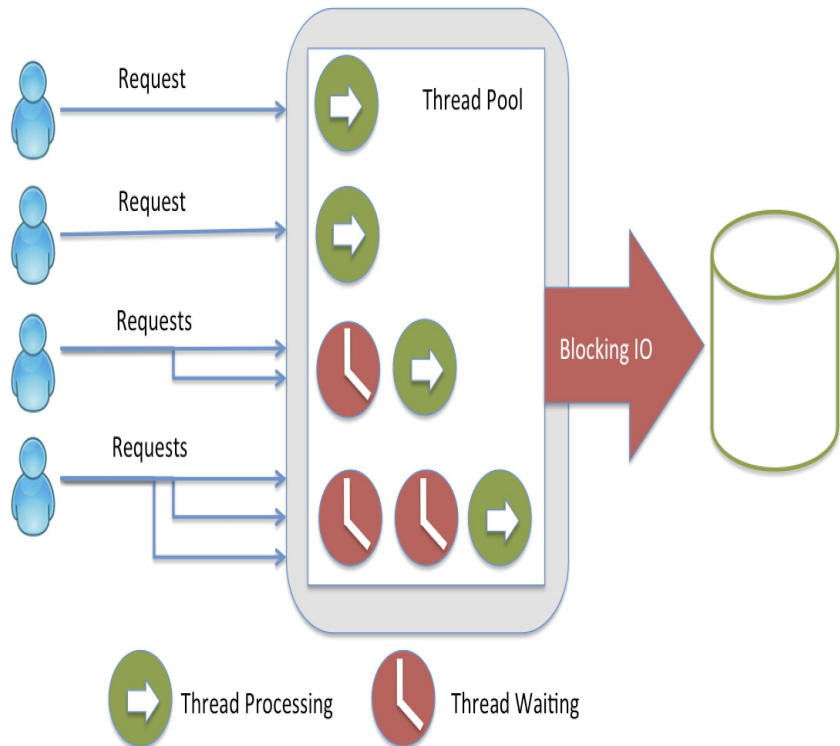
PEARSON



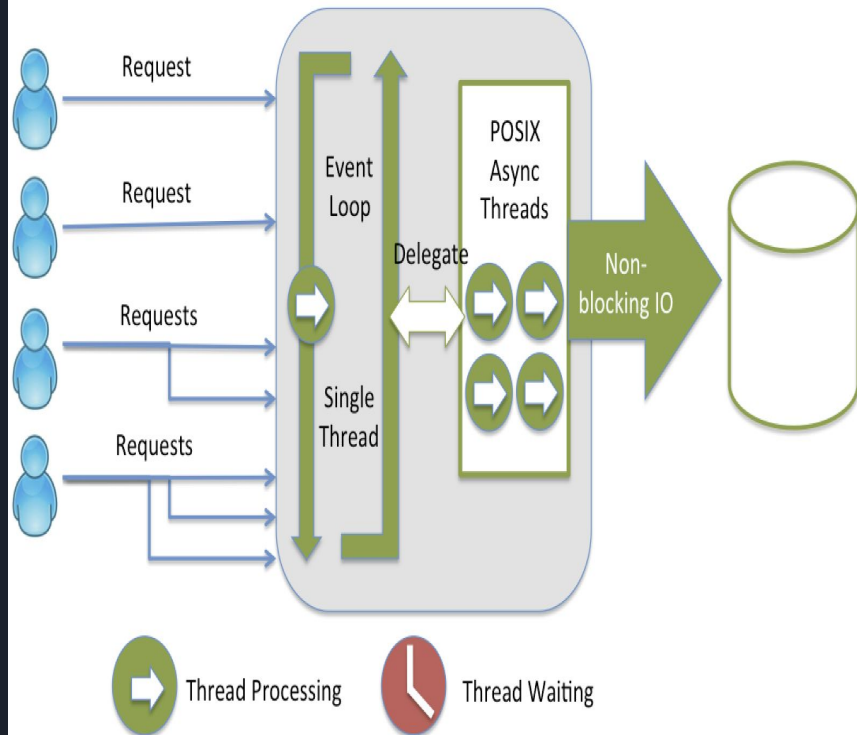
Concurrency VS Parallelism



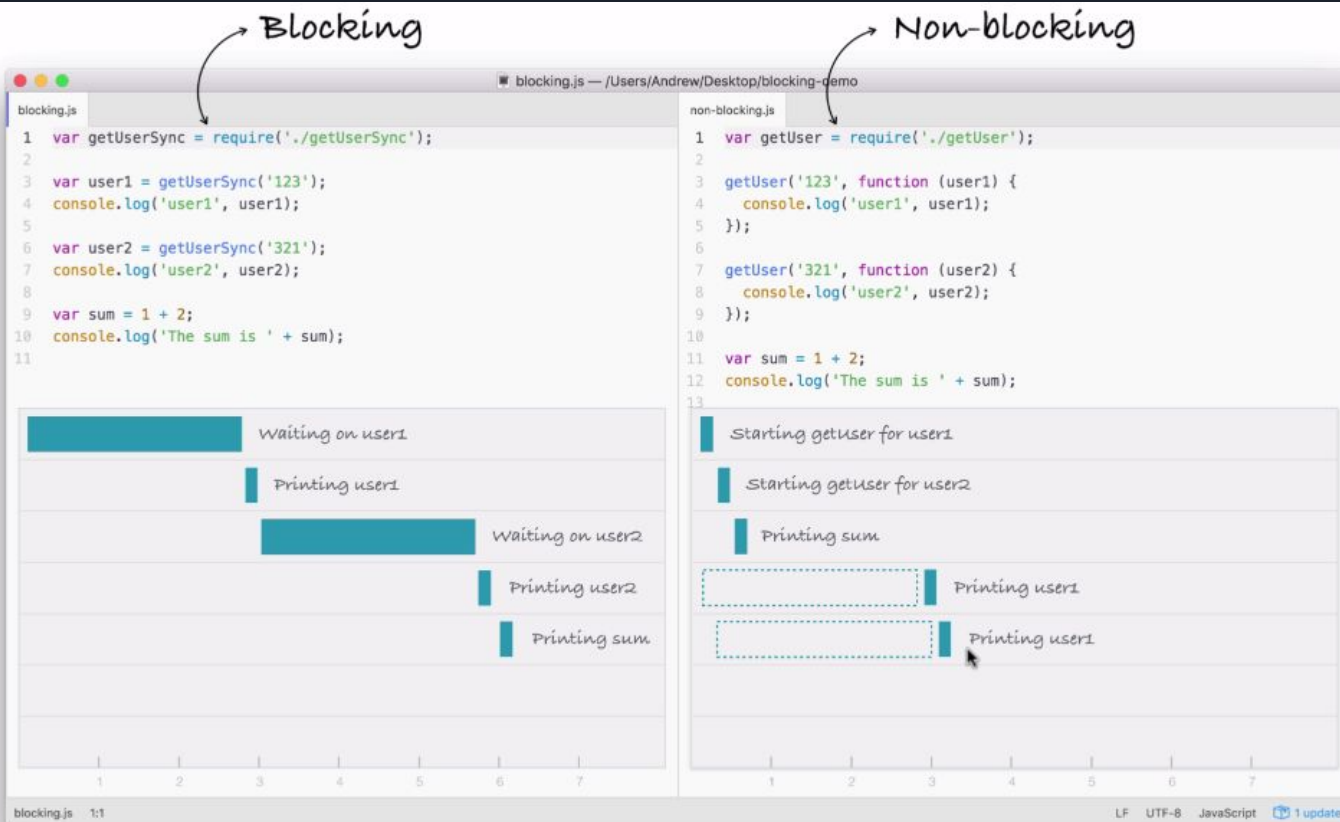
Multi Threaded Server



Node.js Server

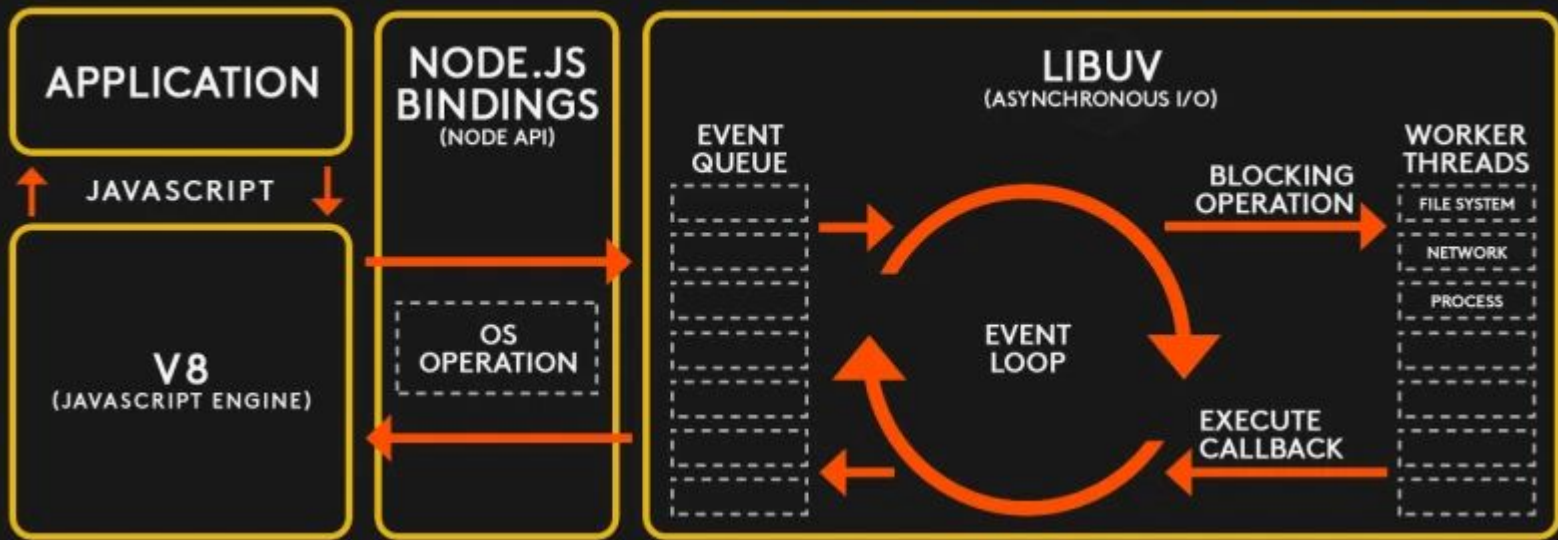


Blocking Vs Non-Blocking Code



THE NODE.JS SYSTEM

A DIAGRAM FROM
MODULUS





JS in browser and Node.js
Node REPL
Node Global Objects
Environment Variables
CLI argv



Core Modules



Modules and require



Building CLI ToDo List

Using fs module



Lab;